OAK INTERNATIONAL SCHOOL

PREMIUM MATHEMATICS ACTIVITY WORKBOOK — CLASS 3

Grade Level: Cinquième (5ème) / IB MYP 3 (Ages 12–13)

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Programmes: IB Middle Years Programme (English) & Beninese National APC Curriculum (Français)

SECTION A — STUDENT INTRODUCTION / INTRODUCTION POUR L'ÉLÈVE

Unlocking Geometric Elegance and Algebraic Foundations

Bienvenue en classe de Cinquième / IB MYP 3 à Oak International School !

In this pivotal academic year, your mathematical universe expands into negative numbers and formal planar properties:

- In **IB MYP 3**, you investigate general algebraic rules (Criterion B), apply scientific notation to astronomical and atomic scales, and construct real-world linear models for distance, speed, and cost (Criterion D).
- In **Beninese 5ème (APC)**, you explore **Situation d'Apprentissage N°2 (Configurations du plan)** and **N°3 (Nombres et calculs)**. You establish the timeless properties of parallel lines cut by a transversal, prove that the angle sum of any triangle is exactly 180^\circ, and uncover the rotational magic of central symmetry (*symétrie centrale*) and parallelograms.

This workbook is your dual-language toolkit to ensure complete mastery of both programmes without cognitive interference.

SECTION B — BILINGUAL MATHEMATICAL LANGUAGE / GLOSSAIRE BILINGUE

Angles, Quadrilaterals & Signed Numbers

English Term	French Equivalent (*Terme Français*)	Mathematical Definition & Symbolic Example	Common Translation Pitfalls
Directed numbers / Integers	*Nombres relatifs / Entiers relatifs*	Signed numbers: \mathbb{Z} = \{\dots, -3, -2, -1, 0, +1, +2, +3, \dots\}.	In French, *relatif* includes both positive and negative values.
Alternate interior angles	*Angles alternes-internes*	Non-adjacent angles on opposite sides of a transversal between parallel lines: \hat{\alpha} = \hat{\beta}.	In French: *alternes-internes* (never say "angles alternatifs").
Corresponding angles	*Angles correspondants*	Angles in matching relative positions at each intersection of two lines with a transversal.	In French: *angles correspondants*.
Vertically opposite angles	*Angles opposés par le sommet*	Angles formed by two intersecting lines sharing a common vertex: \hat{1} = \hat{2}.	In English, often called "vertical angles".
Point reflection / Central symmetry	*Symétrie centrale*	Transformation through point O where O is the midpoint of [MM']: S_O(M) = M'.	In English, equivalent to a rotation of 180^\circ around point O.
Parallelogram	*Parallélogramme*	A quadrilateral with both pairs of opposite sides parallel and equal.	Diagonals bisect each other (*se coupent en leur milieu*).
Scientific notation	*Écriture scientifique*	Expression in form a \times 10^k, where 1 \le a < 10 and k \in \mathbb{Z}.	French *notation scientifique* has the identical mathematical constraint on a.
Order of magnitude	*Ordre de grandeur*	The power of 10 closest to the numerical value: e.g., 3.8 \times 10^5 \approx 10^5 or 10^6.	In French: *ordre de grandeur*.

SECTION C — SHARED CONCEPTUAL FOUNDATIONS

Unit Theme: Parallel Lines, Transversals and Triangle Angle Sums

1. Fundamental Theorem of Angles Cut by a Transversal

Let (d_1) and (d_2) be two parallel lines cut by a secant transversal line (\Delta):

- Any pair of **alternate interior angles** (*angles alternes-internes*) are equal.
- Any pair of **corresponding angles** (*angles correspondants*) are equal.
- Conversely: If two lines cut by a transversal form equal alternate interior angles, then the lines are strictly parallel.

```
(\Delta)
/
-------/------- (d1)
/ \ alpha
/ \
----/-----\---- (d2)
/ beta
/
```

2. Universal Proof: The Angle Sum of a Triangle is 180^\circ

Consider any triangle ABC:

1. Through vertex A, construct auxiliary line (d) parallel to side (BC).
2. Line (d) forms two angles with sides [AB] and [AC]:
• Angle \hat{1} is alternate interior to angle \widehat{ABC}.
• Angle \hat{2} is alternate interior to angle \widehat{ACB}.
3. Angles \hat{1}, \widehat{BAC}, and \hat{2} together form a straight angle on line (d):

\hat{1} + \widehat{BAC} + \hat{2} = 180^\circ

4. Substituting the equal alternate interior angles:

\widehat{ABC} + \widehat{BAC} + \widehat{ACB} = 180^\circ

This proof is identical in both the IB inquiry continuum and the Beninese NPE guide!

3. Worked Example: Signed Numbers and Precedence

Problem: Calculate A = -15 - (-8) + (+12) - (+20).

• **Step 1 (Transform subtractions into additions of opposites)**:

A = -15 + (+8) + (+12) + (-20)

• **Step 2 (Group positive and negative integers)**:

A = [(+8) + (+12)] + [(-15) + (-20)]

A = (+20) + (-35)

• **Step 3 (Calculate final sum)**:

A = -15

SECTION D — CONTEXTUALISED BRIDGING ACTIVITIES

Activity: The Tanguiéta Solar Farm Tracker Alignment

1. Problem Situation

An electrical engineering team is installing dual-axis photovoltaic solar arrays in Tanguiéta (Atacora department).

• Rows of solar panels are mounted along parallel support girders (L_1) and (L_2).
• A sun-tracking mechanical linkage rod (T) crosses girders (L_1) and (L_2) at pivot points P and Q.
• A sensor at pivot P records an interior angle between girder (L_1) and linkage (T) of 64^\circ.
• A robotic actuator at pivot Q must be calibrated to position the lower panel at angle \theta such that the panels remain parallel to prevent shadowing.

2. The IB Perspective (Inquiry into Geometric Models)

• Identify (L_1) and (L_2) as parallel lines.
• Identify linkage (T) as the transversal.
• By geometric angle theorems, the alternate interior angle at Q must equal 64^\circ.
• If the actuator measures the co-interior (interior on the same side) angle \phi:

\phi = 180^\circ - 64^\circ = 116^\circ

3. The Beninese APC Perspective (Structured Proof / *Démonstration*)

• *Données* : Les poutres de support (L_1) et (L_2) sont parallèles : (L_1) \parallel (L_2). La tige sécante (T) coupe (L_1) en P et (L_2) en Q. L'angle \widehat{L_1 P Q} = 64^\circ.
• *Propriété* : Si deux droites parallèles sont coupées par une sécante, alors les angles alternes-internes qu'elles forment ont la même mesure.
• *Calcul et Conclusion* : L'angle alterne-interne \widehat{P Q L_2} et l'angle \widehat{L_1 P Q} sont de même mesure. Donc \theta = 64^\circ.
• *Vérification supplémentaire* : Les angles intérieurs d'un même côté sont supplémentaires : 64^\circ + 116^\circ = 180^\circ.

4. Comparison of Methods

• Both programmes arrive at the exact same angular values (64^\circ and 116^\circ).
• The Beninese APC demands an explicit reference to the statutory property name (*angles alternes-internes*), while the IB emphasizes functional sensor orientation.

5. Student Reflection Question

Why is it physically impossible for solar panels mounted on parallel girders to have unequal alternate interior angles without bending the linkage rod?

Espace de réponse:

_____
_____

_____
_____

6. Application in a New Context

In an urban development in Abomey-Calavi, two parallel avenues are crossed diagonally by a railway line at an angle of 72^\circ. Determine all eight angles formed at the two intersections.

Working space:

_____
_____

_____
_____

SECTION E — PROGRESSIVE PRACTICE / EXERCICES PROGRESSIFS

Level 1: Foundational Understanding (Signed Numbers & Index Laws)

1. Calculate:

a) (-14) + (+9) =

b) (-8) - (-15) =

c) (-6) \times (-7) =

d) (+48) \div (-8) =

2. Write in scientific notation (a \times 10^k):

a) 450\ 000 =

b) 0.000\ 78 =

c) The speed of light: 300\ 000\ 000\text{ m/s} =

d) 125 \times 10^4 =

Level 2: Standard Application (Angles & Fractions with Unlike Denominators)

3. Calculate and give the irreducible fraction:

a) \frac{3}{4} + \frac{5}{6} =

b) \frac{7}{10} - \frac{4}{15} =

c) \frac{5}{8} \times \frac{16}{25} =

4. In triangle KLM, angle \hat{K} = 58^\circ and angle \hat{L} = 74^\circ.

a) Calculate the measure of angle \hat{M}.

b) Is triangle KLM acute, right-angled, or obtuse? Justify.

Level 3: Multi-Step Problems (Central Symmetry & Parallelograms)

5. Let ABCD be a parallelogram with centre O.

a) What is the image of vertex A under the central symmetry S_O?

b) What is the image of segment [AB] under S_O?

c) Prove that line (AB) is parallel to line (CD) using the properties of central symmetry.

6. Proportionality and Journey Rates:

A refrigerated truck carrying fish from Cotonou to Malanville travels 720\text{ km}.

- The truck maintains a constant average speed of 60\text{ km/h}.

• Fuel consumption is directly proportional to distance: 18\text{ litres} per 100\text{ km}.

a) Calculate the total travel time in hours.

b) Calculate the total volume of diesel consumed.

c) If diesel costs 650\text{ CFA F} per litre, calculate the total fuel expenditure.

Level 4: Reasoning & Geometric Proof

7. In the figure below, lines (AB) and (CD) are parallel. Triangle CDE is isosceles at C.

Angle \widehat{BAC} = 45^\circ and angle \widehat{ACD} = 45^\circ.

Line (AC) is a transversal.

Angle \widehat{CDE} = 70^\circ.

a) Prove that triangle ACD is right-angled if \widehat{ADC} = 45^\circ.

b) Calculate the measures of all three interior angles of triangle CDE.

c) Write a structured proof for each deduced angle.

Level 5: Extension & 5ème National Examination Challenge

8. Situation d'Évaluation Type 5ème (SA N°2 & N°4):

Pour sécuriser un domaine agricole à Savè, un géomètre implante quatre bornes M, N, P, Q.

Il vérifie sur le terrain que le point I est le milieu commun des segments [MP] et [NQ].

De plus, il mesure MN = 145\text{ m} et \widehat{MNP} = 65^\circ.

1. Fais une figure géométrique soignée représentant cette situation (échelle 1\text{ cm} pour 20\text{ m}).
2. Démontre que le quadrilatère MNPQ est un parallélogramme.
3. Déduis-en la longueur exacte du côté [PQ] sans nouvelle mesure.
4. Démontre que les droites (MN) et (PQ) sont strictement parallèles.
5. Détermine la mesure de l'angle \widehat{NPQ}.

SECTION F — INTEGRATED APPLICATION TASKS

Task F-1: The Songhaï Bio-Gas Digester Network (MYP Criterion D & Benin SA 5)

Context: The Songhaï Agricultural Innovation Centre in Porto-Novo generates renewable bio-gas from agricultural biomass to power a collective bakery and dairy processing unit.

- The underground main gas pipe runs along straight line (G_1).
- A branch line (G_2) is laid parallel to (G_1).
- A cross-connecting pipe (C) intersects (G_1) at valve V_1 and (G_2) at valve V_2.
- At valve V_1, the connecting pipe forms an acute angle of 52^\circ with line (G_1).
- The pipe length V_1 V_2 = 85\text{ metres}.

Questions:

1. Draw a schematic diagram of the pipe network labelling (G_1), (G_2), and transversal (C).
2. State the measure of the alternate interior angle at valve V_2 and justify using the appropriate theorem.
3. Determine the obtuse angle formed by the cross-connector (C) and line (G_2).
4. Gas flows through the cross-connector at an average speed of 2.5\text{ m/s}. Calculate how many seconds it takes for gas to travel from V_1 to V_2.
5. If the daily gas production is 450\text{ m}^3 and the bakery consumes \frac{5}{9} of this production, calculate the volume of gas remaining for the dairy unit.

SECTION G — PROGRAMME-SPECIFIC REINFORCEMENT

Part G-1: IB MYP 3 Criterion B Pattern Investigation

Investigation: Consecutive Angle Bisector Grids

Consider a sequence of regular polygons:

- Equilateral triangle (3 sides): Interior angle sum = 180^\circ.
- Quadrilateral (4 sides): Interior angle sum = 360^\circ.
- Pentagon (5 sides): Interior angle sum = 540^\circ.

1. Record values in a table for polygons with n = 3, 4, 5, 6, 7, 8 sides.
2. Formulate a general rule for the sum of interior angles S in terms of number of sides n.
3. Test and justify your general rule using triangulation from a single vertex.
4. Calculate the interior angle sum of a polygon with 24 sides (icositetragon).

Part G-2: Beninese NPE Formal Proof Exercise

Exercice de Démonstration Formelle (Classe de 5ème)

Soit un triangle ABC. Soit I le milieu de [AB] et J le milieu de [AC].

Soit K le symétrique de I par rapport au point J.

1. Démontre que le quadrilatère AICK est un parallélogramme.
2. Déduis-en que les segments [CK] et [AI] sont parallèles et de même longueur.
3. Démontre ensuite que le quadrilatère BCKI est un parallélogramme.

Rédige rigoureusement chaque étape avec les propriétés de la symétrie centrale.

SECTION H — ASSESSMENT, REFLECTION & WORKED SOLUTIONS

Diagnostic Pre-Flight Check

- [ ] I can add and subtract negative numbers with confidence.
- [ ] I can find a common denominator for two fractions (e.g., 4 and 6 \to 12).
- [ ] I know that alternate interior angles between parallel lines are equal.
- [ ] I know that the diagonals of a parallelogram bisect each other.

COMPLETE WORKED SOLUTIONS & MARKING SCHEME

Solutions for Section E (Progressive Practice)

1. **Signed Operations** (1 mark each):

a) -14 + 9 = -5

b) -8 + 15 = +7

c) (-6) \times (-7) = +42

d) 48 \div (-8) = -6

2. **Scientific Notation** (1 mark each):

a) 4.5 \times 10^5

b) 7.8 \times 10^{-4}

c) 3.0 \times 10^8\text{ m/s}

d) 1.25 \times 10^6

3. **Fractions Unlike Denominators** (1 mark each):

a) \frac{3 \times 3}{12} + \frac{5 \times 2}{12} = \frac{9+10}{12} = \frac{19}{12}.

b) \frac{7 \times 3}{30} - \frac{4 \times 2}{30} = \frac{21-8}{30} = \frac{13}{30}.

c) \frac{5 \times 16}{8 \times 25} = \frac{1 \times 2}{1 \times 5} = \frac{2}{5}.

4. **Triangle Angles** (2 marks):

a) \hat{M} = 180^\circ - (58^\circ + 74^\circ) = 180^\circ - 132^\circ = 48^\circ.

b) All three angles are strictly less than 90^\circ (58^\circ, 74^\circ, 48^\circ < 90^\circ); therefore, triangle KLM is an acute-angled triangle (*triangle acutangle*).

5. **Parallelogram Central Symmetry** (3 marks):

a) Under S_O, the image of A is C (since O is the midpoint of [AC]).

b) Under S_O, the image of B is D. Thus, the image of segment [AB] is segment [CD].

c) Central symmetry transforms any straight line into a line parallel to itself. Therefore, line (AB) is strictly parallel to line (CD).

6. **Refrigerated Truck** (3 marks):

a) Travel time: $t = \frac{d}{v} = \frac{720}{60} = 12\text{ hours}$.

b) Diesel volume: $\frac{720 \times 18}{100} = 7.2 \times 18 = 129.6\text{ litres}$.

c) Expenditure: $129.6 \times 650 = 84\ 240\text{ CFA F}$.

7. **Angle Reasoning** (4 marks):

a) In $\triangle ACD$, angles are $45^\circ, 45^\circ$. Sum = 90°. Remaining angle $\widehat{ADC} = 180^\circ - 90^\circ = 90^\circ$. Right-angled at D.

b) In $\triangle CDE$ isosceles at C, base angles $\widehat{CDE} = \widehat{CED} = 70^\circ$.

Vertex angle $\widehat{DCE} = 180^\circ - (70^\circ + 70^\circ) = 40^\circ$.

8. **Savè Farm 5ème Task** (5 marks):

1. Scale diagram ($145\text{ m} \to 7.25\text{ cm}$). [1 mark]
2. Segments [MP] and [NQ] have the same midpoint I. A quadrilateral whose diagonals bisect each other is a parallelogram. Thus, MNPQ is a parallelogram. [2 marks]
3. In a parallelogram, opposite sides are equal in length. Hence $PQ = MN = 145\text{ m}$. [1 mark]
4. In a parallelogram, opposite sides are parallel. Hence $(MN) \parallel (PQ)$. [1 mark]

Solutions for Section F (Task F-1: Songhaï Bio-Gas)

1. Labeled schematic diagram. [1 mark]
2. Alternate interior angle at $V_2 = 52^\circ$, because $(G_1) \parallel (G_2)$ cut by transversal (C). [2 marks]
3. Obtuse supplementary angle: $180^\circ - 52^\circ = 128^\circ$. [1 mark]
4. Transit time: $t = \frac{d}{v} = \frac{85}{2.5} = 34\text{ seconds}$. [2 marks]
5. Gas remaining: $\frac{5}{9} \times 450 = 250\text{ m}^3$ used by bakery. Remaining = $450 - 250 = 200\text{ m}^3$. [2 marks]

Solutions for Section G-2 (Benin Formal Proof 5ème)

1. *Démonstration AICK* : J est le milieu de [AC] (par hypothèse) et J est le milieu de [IK] (car K est le symétrique de I par rapport à J). Le quadrilatère AICK a ses diagonales [AC] et [IK] qui se coupent en leur milieu J ; c'est donc un parallélogramme. [2 pts]
2. *Conséquence* : Puisque AICK est un parallélogramme, ses côtés opposés [CK] et [AI] sont parallèles et de même longueur : $(CK) \parallel (AI)$ et $CK = AI$. [1 pt]
3. *Démonstration BCKI* : On sait que I est le milieu de [AB], donc $AI = IB$, et les points A, I, B sont alignés. Comme $CK = AI$, on a aussi $CK = IB$. De plus, comme $(CK) \parallel (AI)$ et que $B \in (AI)$, on a $(CK) \parallel (IB)$. Le quadrilatère BCKI possède deux côtés opposés [CK] et [IB] qui sont parallèles et de même longueur ; c'est donc un parallélogramme. [2 pts]

End of Workbook 3 (Cinquième / MYP 3).